

Assignment – 10.2

2303A51860

Task 1 – Error Detection and Correction

PROMPT :

```
#def calculate_total(nums)
```

```
# sum = 0
```

```
# for n in nums
```

```
# sum += n
```

```
# return total
```

#correct and give brief explanation of all syntax and logical errors.

CODE :

```
1  #def calculate_total(nums)
2  # sum = 0
3  # for n in nums
4  # sum += n
5  # return total
6  #correct and give brief explanation of all syntax and logical errors.
7
8  #Here is the corrected code with explanations of the syntax and logical errors:
9  def calculate_total(nums):
10     total = 0 # Initialize total to 0 instead of sum to avoid confusion with the built-in sum() function
11     for n in nums: # Add a colon at the end of the for loop
12         total += n # Use total instead of sum to accumulate the total
13     return total # Return the total instead of total (which was not defined)
14
15
```

JUSTIFICATION :

This task helps students learn how to identify and fix syntax errors (missing colons, indentation, etc.) and logical errors (wrong variables, incorrect output). It improves debugging skills and teaches how AI can assist in making code executable and correct. It also builds confidence in analyzing faulty programs.

Task 2 – Code Style Standardization

PROMPT :

```
#def findSum(a,b):return a+b
```

```
#print(findSum(5,10))
```

#give Well-structured, consistently formatted Python code following standard style conventions.

CODE :

```

16
17 #def findSum(a,b):return a+b
18 #print(findSum(5,10))
19 #give Well-structured, consistently formatted Python code following standard style conventions.
20 def find_sum(a, b):
21     """Returns the sum of a and b."""
22     return a + b
23
24 print(find_sum(5, 10)) # Call the function and print the result
25
26
27

```

OUTPUT :

```

● PS D:\AI> & 'C:\Users\aatig\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\
cher' '64804' '--' 'D:\AI\AI-10.2.py'
15
○ PS D:\AI>

```

JUSTIFICATION :

This task focuses on writing clean, readable, and professional Python code following standard conventions (PEP-8). It helps students understand the importance of proper formatting, spacing, naming, and structure. Using AI here demonstrates how automated tools can improve code quality and maintainability.

Task 3 – Code Clarity Improvement

PROMPT :

```
#def f(x,y):
```

```
# return x-y*2
```

```
# print(f(10,3))
```

#give Python code rewritten with meaningful function and variable names,proper indentation, and improved clarity.

CODE :

```

28 #def f(x,y):
29 # return x-y*2
30 # print(f(10,3))
31 #give Python code rewritten with meaningful function and variable names,proper indentation, and improved clarity.
32 def calculate_difference(x, y):
33     """Returns the result of x minus twice y."""
34     return x - (y * 2)
35
36 result = calculate_difference(10, 3) # Call the function with meaningful variable names
37 print(result) # Print the result of the calculation
38
39

```

OUTPUT :

```

● PS D:\AI> & 'C:\Users\aatiq\AppData\Local\Programs\Python\Python314\python'
  'D:\AI\AI-10.2.py'
4
○ PS D:\AI>

```

JUSTIFICATION :

This task improves program readability by replacing unclear function and variable names with meaningful ones. It teaches that code should be easy to understand, not just correct. Clear code helps future developers (and yourself) quickly understand logic, reducing errors during maintenance.

Task 4 – Structural Refactoring

PROMPT :

```
print("Hello Ram")
```

```
# print("Hello Sita")
```

```
# print("Hello Ravi")
```

#give Modular Python code using reusable functions to eliminate repetition.

CODE :

```

39
40 #print("Hello Ram")
41 # print("Hello Sita")
42 # print("Hello Ravi")
43 #give Modular Python code using reusable functions to eliminate repetition.
44 def greet(name):
45     """Prints a greeting message for the given name."""
46     print(f"Hello {name}")
47 names = ["Ram", "Sita", "Ravi"] # List of names to greet
48 for name in names: # Loop through the list of names and call the greet function
49     greet(name)
50

```

OUTPUT :

```

● PS D:\AI> & 'C:\Users\aatiq\AppData\Local\Programs\Python\Python314\python'
  'D:\AI\AI-4.2.py'
1
-5
0
○ PS D:\AI>

```

JUSTIFICATION :

This task introduces modular programming by removing repetition and creating reusable

functions. It shows how AI can help refactor duplicated code into cleaner structures, making programs more scalable and organized. It also promotes the DRY principle (Don't Repeat Yourself).

Task 5 – Efficiency Enhancement

PROMPT :

```
numbers = [ ]
```

```
# for i in range(1, 500000):
```

```
# numbers.append(i * i)
```

```
# print(len(numbers))
```

#give Optimized Python code that achieves the same result with improved performance.

CODE :

```
51
52 #numbers = [ ]
53 # for i in range(1, 500000):
54 # numbers.append(i * i)
55 # print(len(numbers))
56 #give Optimized Python code that achieves the same result with improved performance.
57 # Instead of using a loop to append squares to a list, we can use a list comprehension for better performance.
58 numbers = [i * i for i in range(1, 500000)] # Use a list comprehension to generate squares
59 print(len(numbers)) # Print the length of the list of squares
60
61
62
63
```

OUTPUT :

```
● PS D:\AI> & 'C:\Users\aatiq\AppData\Local\Programs\Python\Pyt
' 'D:\AI\AI-10.2.py'
499999
○ PS D:\AI> █
```

JUSTIFICATION :

This task teaches performance optimization by converting slow loops into faster approaches like list comprehensions. It highlights how AI can improve execution speed and memory usage. Students learn to write efficient Python programs, which is important for real-world applications with large data.